

VISHAL GEDELA

Electronics & Communication Engineering Student | VLSI & Digital Design | Embedded Systems | AI/ML
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PROFESSIONAL SUMMARY

Motivated Electronics and Communication Engineering student with a strong interest in VLSI, Digital Design, Embedded Systems, Artificial Intelligence, and Machine Learning. Hands-on experience in Python, Verilog/VHDL, MATLAB, FPGA design, and embedded systems through academic and research projects. Developed an AI-based hand gesture recognition system using IMU sensors and KNN, and a Genetic Algorithm-optimized RF impedance matching system. Passionate about applying engineering and problem-solving skills to real-world technologies and continuously developing expertise in ECE and emerging technologies.

WORK EXPERIENCE

AI Intern | Internship

Jun 2026 - Aug 2026

Waveva Technologies

Remote • Bangalore

- Worked with real-world datasets sourced from **Kaggle** for AI and Machine Learning applications.
- Performed **data cleaning, preprocessing, and exploratory data analysis (EDA)** on datasets.
- Used **Python, Pandas, NumPy, and Matplotlib** for data manipulation and visualization.
- Prepared datasets for machine learning by handling data inconsistencies and selecting relevant features.
- Analyzed dataset patterns and prepared data for model development and evaluation.
- Completed **30+ hours of training** in Artificial Intelligence and Machine Learning.

Technologies / Skills Used : Python, Machine Learning, Pandas, NumPy, Matplotlib, Kaggle, Data Preprocessing, Data Cleaning, Exploratory Data Analysis (EDA)

PROJECTS

Design and Implementation of an AI-Based Hand Gesture Recognition System Using IMU Sensor and KNN Algorithm



Jan 2026 - Apr 2026

Research Project Developer

- Developed a hand gesture recognition system using IMU sensor data and Machine Learning techniques.
- Collected and processed accelerometer and gyroscope data for gesture analysis.
- Implemented the K-Nearest Neighbors (KNN) algorithm for gesture classification.
- Performed data preprocessing, feature extraction, and model training.
- Achieved accurate real-time recognition of predefined hand gestures.
- Demonstrated applications in human-computer interaction and smart control systems.

Technologies / Tools Used : Python, KNN Algorithm, IMU Sensor, NumPy, Scikit-learn

Genetic Algorithm Optimized T-Network Impedance Matching of RF Coils with Complex Biological Loads



Nov 2025 - Jan 2026

Research Project Developer

- Designed an RF impedance matching system for efficient power transfer in biological environments.
- Applied Genetic Algorithms to optimize T-network matching parameters.
- Reduced reflection losses and improved impedance matching performance.
- Simulated and analyzed RF coil behavior under complex biological loading conditions.
- Evaluated system efficiency using optimization and performance metrics.
- Demonstrated the effectiveness of evolutionary algorithms in RF engineering applications.

Technologies / Tools Used : MATLAB, Data Analysis, Genetic Algorithms, RF Circuit Design, Optimization Techniques

CERTIFICATIONS

- Design of CPS with ARM Processor using Embedded C
- Fundamentals of Semiconductor Characterization
- Introduction to Semiconductor Devices 1
- 6G Evolution: Blockchain, Semantic Communications & Radar

SKILLS

Programming Languages :	C, Python, Matlab, Verilog, VHDL
Digital/VLSI :	Digital Logic Design, RTL Design, FPGA
Frameworks & Libraries :	NumPy, Scikit-learn, Pandas, Matplotlib
Tools & Platforms :	GitHub, VS Code, MATLAB, Arduino IDE, Xilinx Vivado, Multisim, LTspice
Soft Skills :	Communication, Leadership, Teamwork, Problem Solving, Time Management
Languages :	English, Telugu, Hindi

PATENTS

- System for **Genetic Algorithm Optimized T-Network Impedance Matching** of RF Coils with **Complex Biological Loads** — *Published Patent Application, Application No. 202641013851*
- **AI-Based Hand Gesture Recognition System** Using IMU Sensor and KNN Algorithm — *Published Patent Application, Application No. 202641018713*

EDUCATION

SRM Institute of Science and Technology | Tiruchirappalli

2024 - 2028

B.Tech — Electronics & Communication Engineering

CGPA : 9.25

- Digital Electronics
- Microprocessors and Microcontrollers
- Signals and Systems
- Artificial Intelligence
- Embedded Systems
- Communication Systems